Theory of Value

An Axiomatic Analysis of Economic Equilibrium
Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

October 1, 2022

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

Continuous Correspondences

a. Example (2) of 1.3.h has pointed out the interest of correspondences for economics; this section will study their continuity.

Continuous Correspondences

b. In this section, S denotes a subset of $\mathbb{R}^K$, T a **compact subset** of $\mathbb{R}^L$, (x^q) a sequence of points of S, (y^q) a sequence of points of T, and a φ correspondence from S to T.

Continuous Correspondences

c. The concept of continuity for a correspondence will be introduced in three steps.

Continuous Correspondences

d. Let x^0 be a point of S; one defines firstly:
The correspondence φ is upper semicontinuous (u.s.c.) at the point x^0 if:

$$\text{“}x^q \to x^0, y^q \in \varphi\left(x^q\right), y^q \to y^0\text{”} \text{ implies } \text{“}y^0 \in \varphi\left(x^0\right).\text{”}$$

In other words, if x^q tends to x^0, and if y^q tends to y^0 while belonging for all q to the image-set of x^q, one requires that y^0 belong to the image-set of x^0. One could also say, if $x^q \to x^0$, and if the distance from y^0 to the image-set of x^q tends to zero, one requires that y^0 belong to the image-set of x^0. This is a natural but rather weak continuity requirement, as the following example shows. Let S and T be two compact real intervals. The graph of φ (fig. 2.a) is the shaded region, heavy-lined boundary included; $\varphi\left(x^0\right)$ is the interval $\left[a^1, a^2\right]$. The correspondence φ is closed at x^0.

Continuous Correspondences

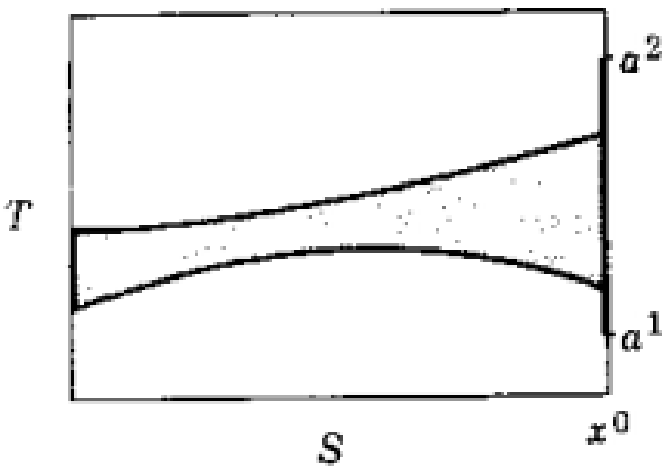


Fig. 2.a

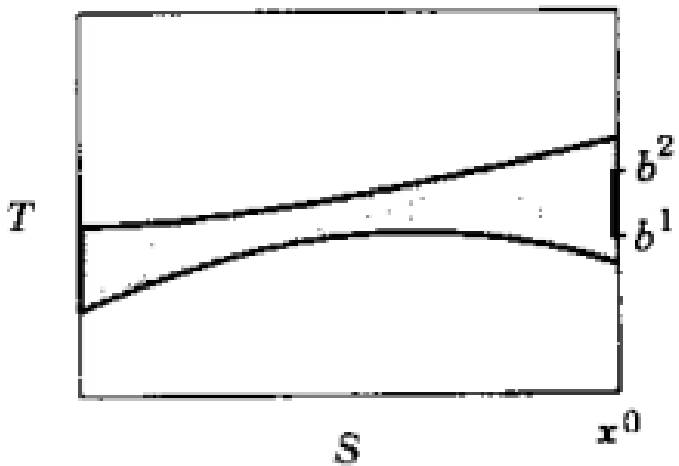


Fig. 2.b

Continuous Correspondences

e. One defines secondly:
The correspondence φ is lower semicontinuous (l.s.c.) at the point x^0 if:

"$x^q \to x^0, y^0 \in \varphi\left(x^0\right)$" implies "there is $\left(y^q\right)$ such that $y^q \in \varphi\left(x^q\right), y^q \to y^0$.

In other words, if x^q tends to x^0, and if y^0 belongs to the image-set of x^0, one requires that there be an infinite sequence $\left(y^q\right)$ such that y^q tends to y^0 while belonging for all q to the image-set of x^q. One could also say, if x^q tends to x^0, and if y^0 belongs to the image-set of x^0, one requires that the distance from y^0 to the image-set of x^q tend to zero. Again, this is a natural but weak continuity requirement as the example of fig. 2.b shows. Here $\varphi\left(x^0\right)$ is the interval $\left[b^1, b^2\right]$. The correspondence φ is lower semi-continuous at x_0. Notice how the roles of "$y^0 \in \varphi\left(x^0\right)$" and "$y^q \in \varphi\left(x^q\right)$, $y^q \to y^0$" are permuted in the two preceding definitions.

Continuous Correspondences

f. Finally one defines:

The correspondence φ is continuous at the point x^0 if it is upper and lower semicontinuous at x^0.

φ then has all the desirable continuity properties at x^0.

Continuous Correspondences

g. When, for all $x \in S$, $\varphi(x)$ consists of a single element, the definition of lower semicontinuity at x^0 is obviously equivalent to the definition of continuity at x^0 for a function. One can prove that, in the same case, the definition of upper semicontinuity at x^0 is equivalent to the definition of continuity at x^0 for a function.

Continuous Correspondences

h. Semicontinuities and continuity on S are defined as semicontinuities and continuity at every point of S. It is clear that:

(1) The correspondence φ is upper semicontinuous on S if and only if its graph is closed in $S \times T$.

Proof. If the graph $G(\varphi)$ is closed in $S \times T$, then the correspondence φ is upper semicontinuous on S. For any $x^0 \in S$, if $x^q \to x^0$, $y^q \in \varphi(x^q)$, and $y^q \to y^0$, then $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$. As $G(\varphi)$ is closed, then $(x^0, y^0) \in G(\varphi)$ and hence $y^0 \in \varphi\left(x^0\right)$.

Proof. If the correspondence φ is upper semicontinuous on S, then its graph is closed in $S \times T$. If $(x^q, y^q) \in G(\varphi)$ and $(x^q, y^q) \to (x^0, y^0)$, then $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q)$. As φ is upper semicontinuous on x^0, then $y^0 \in \varphi\left(x^0\right)$ and hence $(x^0, y^0) \in G(\varphi)$. $\square$

Continuous Correspondences

i. As in 1.7.c let S_1, S_2, T be subsets of $\mathbb{R}^{K_1}$, $\mathbb{R}^{K_2}$, $\mathbb{R}^{L}$ respectively, f be a function from S_1 to S_2, and φ be a correspondence from S_2 to T. Define a correspondence ψ from S_1 to T by $\psi(x) = \varphi(f(x))$ for every x in S_1. It is immediate that:

(2) If f is continuous at the point x of S_1, and φ is upper semicontinuous (resp. lower semicontinuous) at the point $f(x)$ of S_2, then ψ is upper semicontinuous (resp. lower semicontinuous) at x.

Proof. If $x^q \to x$, $y^q \in \psi(x^q)$, and $y^q \to y$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$,
$y^q \in \psi(x^q) = \varphi(f(x^q)) \equiv \varphi(z^q)$. As φ is upper semicontinuous at the point $z \equiv f(x)$, then $y \in \varphi(z) = \varphi(f(x)) = \psi(x)$. Hence ψ is upper semicontinuous at x.

Proof. If $x^q \to x$, and $y \in \varphi(x)$, as$f$ is continuous at the point x, then $z^q \equiv f(x^q) \to f(x) \equiv z$, $y \in \psi(x) = \varphi(f(x)) \equiv \varphi(z)$. As φ is lower semicontinuous at the point $z \equiv f(x)$, then there is (y^q) such that $y^q \in \varphi(z^q) \equiv \varphi(f(x^q)) = \psi(x^q)$ and $y^q \to y$. Hence ψ is lower semicontinuous at x. $\square$

Continuous Correspondences

j. As in 1.7.d-e, for every $n = 1, 2, \cdots, N$ let T_n be a subset of $\mathbb{R}^{L_n}$, and φ_n be a correspondence from S to T_n. Consider the product $T = \prod_{n=1}^{N} T_n$ and define a correspondence φ from S to T by $\varphi(x) = \prod_{n=1}^{N} \varphi_n(x)$ for every x in S. If every T_n is compact, so is T by (7) of 1.6.t. In this case it is immediate that:

(3) If every φ_n is upper semicontinuous (resp. lower semicontinuous) at x of S, then φ is upper semicontinuous (resp. lower semicontinuous) at x.

Continuous Correspondences

Proof. If $x^q \to x$, $y^q \in \varphi(x^q)$, and $y^q \to y$, then $(y_1^q, \cdots, y_N^q) = y^q \in \varphi(x^q) = \prod_{n=1}^{N} \varphi_n(x^q)$ and $(y_1^q, \cdots, y_N^q) = y^q \to y = (y_1, \cdots, y_N)$, that is, $y_n^q \in \varphi_n(x^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. As φ_n is upper semicontinuous at x, then $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. Therefore $y = (y_1, \cdots, y_N) \in \prod_{n=1}^{N} \varphi_n(x) = \varphi(x)$. Hence φ is upper semicontinuous at x.

Proof. If $x^q \to x$, and $y \in \varphi(x)$, then $(y_1, \cdots, y_N) = y \in \varphi(x) = \prod_{n=1}^{N} \varphi_n(x)$, that is, $y_n \in \varphi_n(x)$ for $n = 1, \cdots, N$. As φ_n is lower semicontinuous at the point x, then there is (y_n^q) such that $y_n^q \in \varphi_n(z^q)$ and $y_n^q \to y_n$ for $n = 1, \cdots, N$. Thus $y^q = (y_1^q, \cdots, y_N^q) \in \prod_{n=1}^{N} \varphi_n(x^q) = \varphi(x^q)$ and $y^q = (y_1^q, \cdots, y_N^q) \to (y_1, \cdots, y_N) = y$. Hence φ is lower semicontinuous at x. $\square$

Continuous Correspondences

k. According to example (2) of 1.3.h. the interest of these concepts for economics lies, in particular, in the interpretations of an element x of S as the environment of a certain agent, of T as the set of actions a priori available to him, and of $\varphi(x)$ (assumed here to be closed for every x in S) as the subset of T to which his choice is actually restricted by his environment x. Let f be a continuous real-valued function on $S \times T$, and interpret $f(x, y)$ as the gain for that agent when his environment is x and his action y. Given x, one is interested in the elements of $\varphi(x)$ which maximize f (now a function of y alone) on $\varphi(x)$; they form a set $\mu(x)$. What can be said about the continuity of the correspondence μ from S to T?

$$\mu(x) = \{y \in \varphi(x) \mid y \text{ maximizes } f(x, y)\}$$

Continuous Correspondences

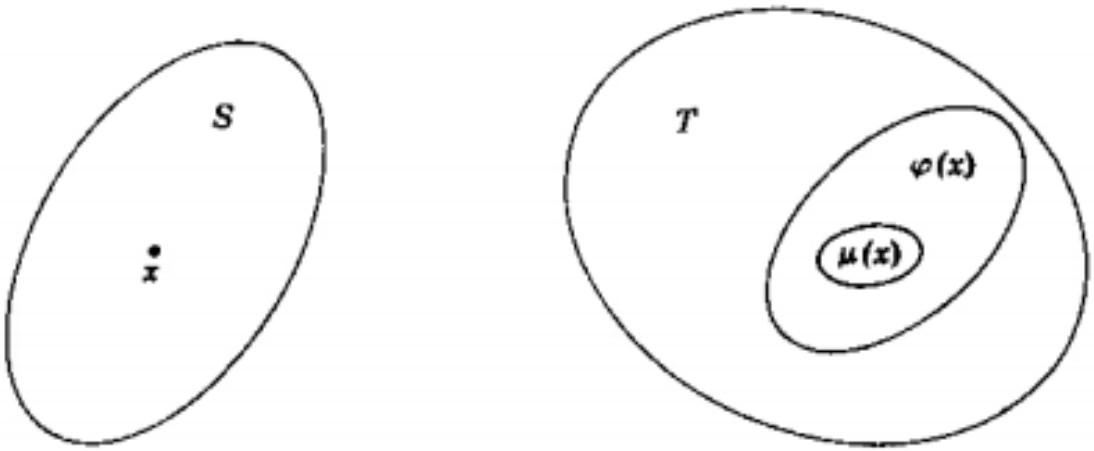


Fig. 3

Continuous Correspondences

One is also interested in $g(x)$, the value of the maximum of f on $\varphi(x)$ for a given x.

$$g(x) = \max\{f(x,y) \mid y \in \varphi(x)\}$$

What can be said about the continuity of the real-valued function g on S? An answer to these two questions is given by the following result (the proof of continuity of g should not be attempted).

(4) If f is continuous on $S \times T$, and if φ is continuous at $x \in S$, then μ is upper semicontinuous at x, and g is continuous at x.

Continuous Correspondences

(4) If f is continuous on $S \times T$, and if φ is compact-valued continuous at $x_0 \in S$, then μ is upper semicontinuous at x_0, and g is continuous at x_0.

(4) If f is continuous on $S \times T$, and if φ is compact-valued (lower semi-continuous and upper semi-continuous) continuous at $x_0 \in S$, define the optimal value function $g : S \to \mathbb{R}$,

$$g(x) = \max \{ f(x, y) \mid y \in \varphi(x) \},$$

and argmax correspondence $\mu : S \to T$,

$$\begin{aligned} \mu(x) &= \{ y \in \varphi(x) \mid y \text{ maximizes } f(x, y) \} \\ &= \{ y \in \varphi(x) \mid f(x, y) = g(x) \} \end{aligned}$$

then g is continuous at x_0, and μ is compact-valued upper semicontinuous at x_0.

Continuous Correspondences

Proof. Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is lower semi-continuous continuous (l.s.c.) at $x_0 \in S$, for any $y_0 \in \varphi(x_0)$, there exists $(y_n) \subseteq T$ such that $y_n \in \varphi(x_n)$ and $y_n \to y^0$. By the continuity of f, we have $f(x_n, y_n) \to f(x_0, y_0)$. Since $f(x_n, y_n) \leqq g(x_n)$, then

$$f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \leqq \underline{\lim}_{n\to\infty} g(x_n)$$

and hence $g(x_0) \leqq \underline{\lim}_{n\to\infty} g(x_n)$. Therefore g is lower semi-continuous continuous (l.s.c.) at x_0.

Fix $x_0 \in S$ and let $x_n \to x_0$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, for any sequence $(y_n \in \varphi(x_n))$, the exists a subsequence $(y_{n_k} \in \varphi(x_{n_k}))$ such that $y_{n_k} \to y_0$ and $y^0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then

$$g(x_0) = f(x_0, y_0) = \lim_{n\to\infty} f(x_n, y_n) \geqq \overline{\lim}_{n\to\infty} g(x_n).$$

Therefore g is upper semi-continuous continuous (u.s.c.) at x_0.

Continuous Correspondences

Proof. For each x_0, f is continuous on $S \times T$, and if φ is compact-valued continuous at $x_0 \in S$, then $\mu(x_0)$ is non-empty and compact-valued from the Extreme Value Theorem.

Let $x_n \to x_0$, $y_n \in \mu(x_n)$. Since φ is compact-valued upper semi-continuous continuous (u.s.c.) at $x_0 \in S$, then there exists $(y_{n_k}) \subseteq T$ such that $y_{n_k} \in \varphi(x_{n_k})$ and $y_{n_k} \to y_0$ and $y_0 \in \varphi(x_0)$. By the continuity of f, we have $f(x_{n_k}, y_{n_k}) \to f(x_0, y_0)$. By the continuity of g, we have $g(x_{n_k}) \to g(x_0)$. Since $f(x_{n_k}, y_{n_k}) = g(x_{n_k})$, then $f(x_0, y_0) = g(x_0)$, so $y_0 \in \mu(x_0)$. Thus μ is upper semicontinuous (u.s.c.) at x_0. $\square$

Continuous Correspondences

Example: Let S and T be the real interval $[0,1]$. Define φ by $\varphi(x) = [0,1]$ for all $x \in S$, and f by $f(x,y) = xy$. For $x \neq 0$, $\mu(x)$ consists of the single element 1; for $x = 0$, $\mu(x) = [0,1]$.

$$\mu(x) = \begin{cases} [0,1], & \text{for} \quad x = 0, \\ \{1\}, & \text{for} \quad x \neq 0. \end{cases}$$

(It will be helpful to draw the graph of μ.) For all x, $g(x) = x$.

$$g(x) = \max\{xy \mid y \in [0,1]\} = x \max\{y \mid y \in [0,1]\} = x.$$

Continuous Correspondences

I. Throughout this section, the assumption that T is compact has been made; it is essential in several respects. In applications, when the set T is not compact, one may still be able to replace T by some compact set without changing the problem, and thus to use the above results.

Semi-continuous Mappings

In this section, S denotes a subset of $\mathbb{R}^K$, T a subset of $\mathbb{R}^L$, and a φ correspondence from S to T, that is $\varphi : S \rightrightarrows T$. For correspondences there are two natural notions of inverse.

(1) The lower (or weak) inverse of $E \subseteq T$ under $\varphi : S \rightrightarrows T$, denoted $\varphi^-(E)$, is defined by

$$\varphi^-(E) = \{x \in S : \varphi(x) \cap E \neq \varnothing\}.$$

(2) The upper (or strong) inverse of $E \subseteq T$ under $\varphi : S \rightrightarrows T$, denoted $\varphi^+(E)$, is defined by

$$\varphi^+(E) = \{x \in S : \varphi(x) \subseteq E\}.$$

Remark. For a single-valued mapping $\varphi : S \rightrightarrows T$, define

$$\varphi^{-1}(y) = \{x \in S : y \in \varphi(x)\} = \{x \in S : y = \varphi(x)\}.$$

Note that $\varphi^{-1}(y) = \varphi^-(\{y\})$.

Semi-continuous Mappings

Let φ be a correspondence of a topological space S into a topological space T and let x_0 be a point of S.

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$ there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

  i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$.

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$ there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

  i.e. $\varphi(U(x_0)) \subseteq G$.

We say that the mapping φ is continuous at x_0 if it is both lower and upper semi-continuous at x_0.

$$\varphi^-(E) = \{x \in S \mid \varphi(x) \cap E \neq \varnothing\}$$
$$\varphi^+(E) = \{x \in S \mid \varphi(x) \subseteq E\}$$

Semi-continuous Mappings

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$ there exists a neighbourhood $U(x_0)$ such that $\varphi(U(x_0)) \cap G \neq \varnothing$

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing.$$

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[x_0 \in \varphi^-(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^-(G)$

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^-(G).$$

- We say that φ is lower semi-continuous (l.s.c.) at x_0 if for each open set G meeting $\varphi(x_0)$ $[x_0 \in \varphi^-(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^-(G)$.
- We say that φ is lower semi-continuous (l.s.c.) at x_0 if $\varphi^-(G)$ is a open set in S for each open set G (meeting $\varphi(x_0)$).

Semi-continuous Mappings

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$ there exists a neighbourhood $U(x_0)$ such that $\varphi(U(x_0)) \subseteq G$

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G.$$

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[x_0 \in \varphi^+(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G).$$

- We say that φ is upper semi-continuous (u.s.c.) at x_0 if for each open set G containing $\varphi(x_0)$ $[x_0 \in \varphi^+(G)]$ there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$.
- We say that φ is upper semi-continuous (u.s.c.) at x_0 if $\varphi^+(G)$ is a open set in S for each open set G (containing $\varphi(x_0)$).

Semi-continuous Mappings

If φ is a single-valued mapping, the definition given above for lower semi-continuity coincides with the ordinary definition of continuity; the same is true for upper semi-continuity.

- We say that φ is lower semi-continuous (l.s.c.) in S if it is lower semi-continuous (l.s.c.) at each point of S, that is, $\varphi^-(G)$ is a open set in S for each open set G.
- We say that φ is upper semi-continuous (u.s.c.) in S if it is upper semi-continuous (u.s.c.) at each point of S, that is, $\varphi^+(G)$ is a open set in S for each open set G.

We say that the correspondence φ is continuous in S if it is both lower and upper semi-continuous in S.

Semi-continuous Mappings

Theorem 1.1. $\varphi : S \rightrightarrows T$ is a correspondence, then the following three statements are equivalent.

(1) φ is lower semi-continuous (l.s.c.).

(2) $\varphi^{-}(G)$ is a open set in S for any open set $G \subseteq T$.

(3) $\varphi^{+}(F)$ is a closed set in S for any closed set $F \subseteq T$.

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^{-}(G)$ is a open set in S. If $x_0 \in \varphi^{-}(G)$, then $\varphi(x_0) \cap G \neq \varnothing$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

i.e. $U(x_0) \subseteq \varphi^{-}(G)$. Therefore $\varphi^{-}(G)$ is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^{-}(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^{-}(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^{-}(G)$, that is, if G meets $\varphi(x_0)$ $[\varphi(x_0) \cap G \neq \varnothing]$, then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^{-}(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \cap G \neq \varnothing,$$

i.e. $\varphi(U(x_0)) \cap G \neq \varnothing$, thus φ is lower semi-continuous (l.s.c.) at x_0.

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq F\} \\
&= \mathbf{C}_S\left(\varphi^+(F)\right)
\end{aligned}$$

is open, that is, $\varphi^+(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T G\} \\
&= \{x \in S \mid \varphi(x) \cap G = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^-(G)$ is open. $\square$

Semi-continuous Mappings

Theorem 1.2. $\varphi : S \rightrightarrows T$ is a correspondence, then the following three statements are equivalent.

(1) φ is upper semi-continuous (u.s.c.).

(2) $\varphi^+(G)$ is a open set in S for any open set $G \subseteq T$.

(3) $\varphi^-(F)$ is a closed set in S for any closed set $F \subseteq T$.

Semi-continuous Mappings

Proof of (1) $\Longrightarrow$ (2) For any open set $G \subseteq T$ we want to prove that $\varphi^+(G)$ is a open set in S. If $x_0 \in \varphi^+(G)$, then $\varphi(x_0) \subseteq G$, and so there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, whence

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

i.e. $U(x_0) \subseteq \varphi^+(G)$. Therefore $\varphi^+(G)$ is a neighbourhood of each of its points and so is open.

Proof of (2) $\Longrightarrow$ (1) Suppose now that $\varphi^+(G)$ is an open set for each open set G in T. If $x_0 \in \varphi^+(G)$, then there exists a neighbourhood $U(x_0)$ such that $U(x_0) \subseteq \varphi^+(G)$, that is, if G contains $\varphi(x_0)$ $[\varphi(x_0) \subseteq G]$, then there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad x \in \varphi^+(G),$$

hence

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G,$$

i.e. $\varphi(U(x_0)) \subseteq G$, thus φ is upper semi-continuous (u.s.c.) at x_0.

Semi-continuous Mappings

Proof of (2) $\Longrightarrow$ (3) For any closed set $F \subseteq T$, $G = \mathbf{C}_T F$ is an open set.

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\left(\varphi^-(F)\right)
\end{aligned}$$

is open, that is, $\varphi^-(F)$ is closed.

Proof of (3) $\Longrightarrow$ (2) For any open set $G \subseteq T$, $F = \mathbf{C}_T G$ is a closed set.

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \cap F = \varnothing\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq \mathbf{C}_T F\} \\
&= \mathbf{C}_S\{x \in S \mid \varphi(x) \subseteq G\} \\
&= \mathbf{C}_S\left(\varphi^+(G)\right)
\end{aligned}$$

is closed, that is, $\varphi^+(G)$ is open. $\square$

Semi-continuous Mappings

Upper semi-continuous (u.s.c.) correspondences of S into T have the following fundamental property:

Theorem 1.3. If $\varphi : S \to T$ is a compact-valued u.s.c. correspondence, then the image $\varphi(K)$ of a compact subset K of S is also compact.

Proof. Let $\{G_\lambda \mid \lambda \in \Lambda\}$ be an open covering of $\varphi(K)$. If $x \in K$, the set $\varphi(x)$, which is compact, can be covered by a finite number of the $G_\lambda(x)$ for $\lambda \in \Lambda$, $\varphi(x) \subseteq \bigcup_{\lambda\in\Lambda} G_\lambda(x)$; let G_x denote the union of the sets in such a finite family, $G_x = \bigcup_{\lambda\in\Lambda} G_\lambda(x)$, then $\varphi(K) \subseteq \bigcup_{x\in K} \varphi(x) \subseteq \bigcup_{x\in K} G_x$. Then $\{\varphi^+(G_x) \mid x \in K\}$ is an open covering of K and so it contains a finite covering $\varphi^+(G_{x_1}), \cdots, \varphi^+(G_{x_n}), \cdots, \varphi^+(G_{x_N})$. $K \subseteq \bigcup_{n=1}^{N} \varphi^+(G_{x_n})$. The sets $G_{x_1}, \cdots, G_{x_n}, \cdots, G_{x_N}$ cover $\varphi(K)$ and so $\varphi(K)$ can be covered by a finite number of the G_{x_n} for $n = 1, \cdots, N$. That is, $\varphi(K) \subseteq \bigcup_{n=1}^{N} G_{x_n}$.

□

Semi-continuous Mappings

In addition to the two types of semi-continuity, it is sometimes convenient to consider a third topological property.

Definition 1.2. (Closed Correspondence) We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.

Consider the graphical representation $\sum_{x \in S} \varphi(x)$ of φ in $S \times T$;

$$G(\varphi) \equiv \sum_{x \in S} \varphi(x) = \{(x, y) \in S \times T \mid y \in \varphi(x)\}.$$

Semi-continuous Mappings

- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ and $y \in V(y_0)$ implies $y \notin \varphi(x)$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $(x_0, y_0) \in \mathbf{C}(G(\varphi))$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ and $y \in V(y_0)$ implies $(x, y) \in \mathbf{C}(G(\varphi))$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $(x_0, y_0) \in \mathbf{C}(G(\varphi))$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $U(x_0) \times V(y_0) \subseteq \mathbf{C}(G(\varphi))$.
- We say that φ is a closed correspondence of S into T if $\mathbf{C}(G(\varphi))$ is an open set in $S \times T$.
- We say that φ is a closed correspondence of S into T if $G(\varphi)$ is a closed set in $S \times T$.

Semi-continuous Mappings

Proposition 1.1. Let S and T be two metric spaces. A mapping $\varphi : S \rightrightarrows T$ is a closed correspondence if and only if $G(\varphi)$ is a closed set in $S \times T$.

Semi-continuous Mappings

- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, $y_0 \notin \varphi(x_0)$ there exist two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \varnothing$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, for any two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) \neq \varnothing$ [$y \in \varphi(x) \cap V(y_0)$ if and only if $y \in \varphi(x)$ and $y \in V(y_0)$], then $y_0 \in \varphi(x_0)$.
- We say that φ is a closed correspondence of S into T if whenever $x_0 \in S$, $y_0 \in T$, for any two neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$, $y \in V(y_0)$, and $y \in \varphi(x)$, then $y_0 \in \varphi(x_0)$.
- We say that φ is a closed correspondence of S into T if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Semi-continuous Mappings

Proposition 1.2. Let S and T be two metric spaces. A mapping $\varphi : S \rightrightarrows T$ is a closed correspondence if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Theorem 1.4. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$.

Proof. $G(\varphi)$ is a closed set in $S \times T$ if and only if whenever $(x_n, y_n) \longrightarrow (x_0, y_0)$ and $(x_n, y_n) \in G(\varphi)$, then $(x_0, y_0) \in G(\varphi)$ if and only if whenever $x_n \longrightarrow x_0$, $y_n \longrightarrow y_0$, $y_n \in \varphi(x_n)$, then $y_0 \in \varphi(x_0)$. $\square$

Properties of Semi-continuous Mappings

Proposition 1.3. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n : S \rightrightarrows \prod_{n=1}^{N} T_n$ is a closed correspondence.

Proof 1. If $\left(y_1^0, \cdots, y_N^0\right) = y^0 \notin \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then there exists $n_0 = 1, \cdots, N$ such that $y_{n_0}^0 \notin \varphi_{n_0}\left(x^0\right)$. Therefore there exist neighbourhoods $U_{n_0}\left(x^0\right)$ and $V_{n_0}\left(y_{n_0}^0\right)$ such that $x \in U_{n_0}\left(x^0\right)$ implies $\varphi_{n_0}(x) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$, that is, $\varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $U\left(x^0\right) = \bigcap_{n=1}^{N} U_n\left(x^0\right)$, then $x \in U(x_0)$ implies

$$\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) \subseteq \varphi_{n_0}\left(U_{n_0}\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$$

and hence $\varphi_{n_0}\left(U\left(x^0\right)\right) \cap V_{n_0}\left(y_{n_0}^0\right) = \varnothing$. Take $V\left(y^0\right) = \prod_{n=1}^{N} V_n\left(y_n^0\right)$, then

$$\varphi\left(U\left(x^0\right)\right) \cap V\left(y^0\right) = \prod_{n=1}^{N}\left[\varphi_n\left(U\left(x^0\right)\right) \cap V_n\left(y_n^0\right)\right] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $\left(y_1^q, \cdots, y_N^q\right) = y^q \to y^0 = \left(y_1^0, \cdots, y_N^0\right)$, and $\left(y_1^q, \cdots, y_N^q\right) = y^q \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$, then $y_n^q \to y_n^0$ and $y_n^q \in \varphi_n\left(x^q\right)$ for $n = 1, \cdots, N$. If every φ_n is closed at the point x_0 of S, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$, and hence $y^0 = \left(y_1^0, \cdots, y_N^0\right) \in \prod_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \prod_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

Semi-continuous Mappings

Proposition 1.4. Let S and T be two metric spaces. If $\varphi : S \rightrightarrows T$ is a closed correspondence at x, then the set $\varphi(x)$ is closed in T.

Proof: Let y_n be a sequence in $\varphi(x)$ with $y_n \to y$. Define $x_n = x$ for all n, so $x_n \to x$ and $y_n \in \varphi(x_n)$. Since φ is a closed correspondence at x, we have $y \in \varphi(x)$. This shows that $\varphi(x)$ is a closed set. $\square$

Semi-continuous Mappings

Example 1.1. If f is a continuous numerical function in $S \times T$, the correspondence $\varphi : S \rightrightarrows T$ defined by $\varphi(x) = \{y \in T \mid f(x, y) \leqq 0\}$ is a closed correspondence of S into T, for, in $S \times T$, the graphical representation

$$G(\varphi) = \{(x, y) \in S \times T \mid f(x, y) \leqq 0\}$$

is a closed set.

In particular, if λ is a continuous numerical function in a metric space (X, d), the mapping

$$\varphi(x) = B_{\lambda(x)}(x) = \{y \in X \mid d(x, y) - \lambda(x) \leqq 0\}$$

is closed.

Semi-continuous Mappings

Theorem 1.5. If $\{\varphi_\lambda \mid \lambda \in \Lambda\}$ is a family of closed correspondences of S into T, then $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a closed correspondence.

Proof 1. If $y_0 \notin \varphi(x_0) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0)$, then there exists an index $\lambda_0 \in$ such that

$$y_0 \notin \varphi_{\lambda_0}(x_0).$$

Therefore there exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi_{\lambda_0}(x) \cap V(y_0) = \varnothing$, that is,

$$\varphi_{\lambda_0}(U(x_0)) \cap V(y_0) = \varnothing$$

and so $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \bigcap_{\lambda\in\Lambda} [\varphi_\lambda(x) \cap V(y_0)] = \varnothing$, that is,

$$\varphi(U(x_0)) \cap V(y_0) = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x_n \to x_0$, $y_n \to y_0$, and $y_n \in \varphi(x_n) = \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_n)$, then $y_n \in \varphi_\lambda(x_n)$ for $\lambda \in \Lambda$. If every φ_λ is closed at the point x_0 of S, then $y_0 \in \varphi_\lambda(x_0)$ for $\lambda \in \Lambda$, and hence $y_0 \in \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x_0) = \varphi(x_0)$. Thus, $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is closed at the point x_0 of S.

□

Properties of Semi-continuous Mappings

Proposition 1.5. If every $\varphi_n : S \rightrightarrows T_n$ is a closed correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is a closed correspondence.

Proof 1. If $y_0 \notin \varphi(x_0) = \bigcup_{n=1}^{N} \varphi_n(x_0)$, then, for all $n = 1, \cdots, N$,

$$y_0 \notin \varphi_n(x_0).$$

Therefore there exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that $x \in U(x_0)$ implies $\varphi_n(x) \cap V(y_0) = \varnothing$, that is,

$$\varphi_n(U(x_0)) \cap V(y_0) = \varnothing$$

and so $x \in U(x_0)$ implies $\varphi(x) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(x) \cap V(y_0)] = \varnothing$, that is,

$$\varphi(U(x_0)) \cap V(y_0) = \bigcup_{n=1}^{N} [\varphi_n(U(x_0)) \cap V(y_0)] = \varnothing;$$

whence φ is a closed mapping. □

Proof 2. If $x^q \to x^0$, $y^q \to y^0$, and $y^q \in \varphi(x^q) = \bigcup_{n=1}^{N} \varphi_n(x^q)$, then there exists $n_0 \in \{1, \cdots, N\}$ such that $y^q \in \varphi_{n_0}(x^q)$. If every φ_n is closed at the point x_0 of S, then $y^0 \in \varphi_{n_0}\left(x^0\right)$, and hence $y^0 \in \bigcup_{n=1}^{N} \varphi_n\left(x^0\right) = \varphi\left(x^0\right)$. Thus, $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is closed at the point x_0 of S. □

Semi-continuous Mappings

Theorem 1.6. Every compact-valued u.s.c. correspondence is closed.

Proof. Let φ be a upper semi-continuous (u.s.c.) correspondence of S into T and suppose that $y_0 \notin \varphi(x_0)$. Since $\varphi(x_0)$ is compact, there exists an open set G in T containing $\varphi(x_0)$ and a neighbourhood $V(y_0)$ such that

$$G \cap V(y_0) = \varnothing.$$

Since φ is a upper semi-continuous (u.s.c.) correspondence, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G.$$

Then

$$x \in U(x_0) \Longrightarrow \varphi(x) \cap V(y_0) = \varnothing$$

and φ is therefore closed. $\square$

Semi-continuous Mappings

Theorem 1.7. If φ_1 is a closed mapping of S into T and φ_2 is a compact-valued u.s.c. correspondence of S into T, then the correspondence $\varphi = \varphi_1 \cap \varphi_2$ is compact-valued u.s.c.

Semi-continuous Mappings

Proof. For each x, the set $\varphi(x)$ is compact, because it is a closed set contained in a compact set $\varphi_2(x)$. Thus φ is compact-valued.
Let G be an open set such that $\varphi(x_0) = \varphi_1(x_0) \cap \varphi_2(x_0) \subseteq G$. We shall prove that there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi(x) \subseteq G$$

that is, $\varphi(U(x_0)) \subseteq G$.
(1) If $\varphi_2(x_0) \subseteq G$, there is nothing to prove. φ_2 is upper semi-continuous (u.s.c.), there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow \varphi_2(x) \subseteq G$$

that is, $\varphi_2(U(x_0)) \subseteq G$ and hence $\varphi(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G$.
(2) Suppose that $\varphi_2(x_0) \cap (\mathbf{C}G) = K \neq \varnothing$. Let y be a point of K, $y \notin \varphi_1(x_0)$; then there exist neighbourhoods $V(y)$ and $U_y(x_0)$ such that

$$x \in U_y(x_0) \Longrightarrow \varphi_1(x) \cap V(y) = \varnothing,$$

that is, $\varphi_1(U_y(x_0)) \cap V(y) = \varnothing$.

Semi-continuous Mappings

Since the set K is compact, there exist elements $y_1, y_2, \cdots, y_N$ in K such that $V(y_1), V(y_2), \cdots, V(y_N)$ cover K. Write $K \subseteq \bigcup_{n=1}^{N} V(y_n) \equiv V(K)$, then $\varphi_2(x_0) \subseteq G \cup V(K)$.
If φ_2 is upper semi-continuous (u.s.c.), then there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi_2(x) \in G \cup V(K),$$

that is, $\varphi_2(U'(x_0)) \subseteq G \cup V(K)$.
Putting $U(x_0) = U_{y_1}(x_0) \cap U_{y_2}(x_0) \cap \cdots \cap U_{y_N}(x_0) \cap U'(x_0)$, we have

$$\varphi_1(U(x_0)) \cap V(y_n) \subseteq \varphi_1(U_{y_n}(x_0)) \cap V(y_n) = \varnothing,$$

that is, $\varphi_1(U(x_0)) \cap V(y_n) = \varnothing$, and hence

$$\varphi_1(U(x_0)) \cap V(K) = \varphi_1(U(x_0)) \cap \bigcup_{n=1}^{N} V(y_n) = \bigcup_{n=1}^{N} [\varphi_1(U(x_0)) \cap V(y_n)] = \varnothing$$

Semi-continuous Mappings

Therefore

$$\begin{cases} (\varphi_1 \cap \varphi_2)(U(x_0)) \cap V(K) \subseteq \varphi_1(U(x_0)) \cap V(K) = \varnothing, \\ (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq \varphi_2(U(x_0)) \subseteq G \cup V(K), \end{cases}$$

whence

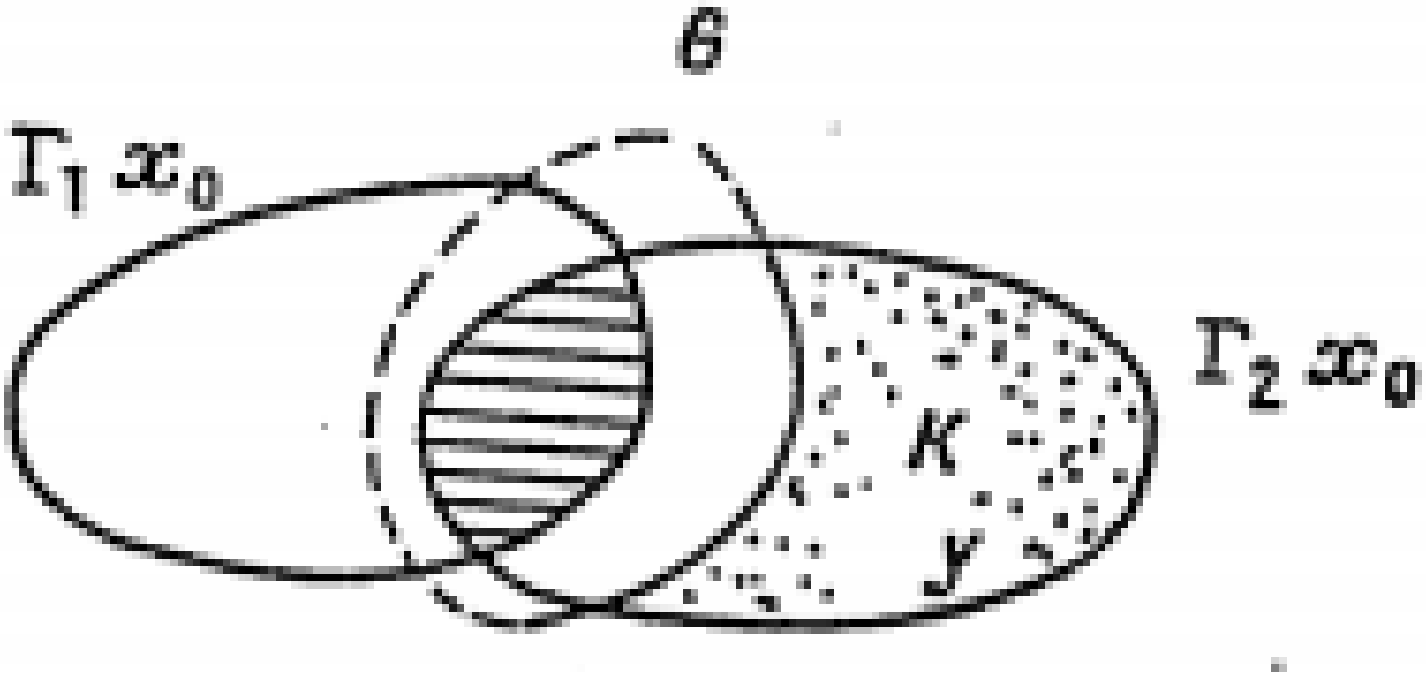
$$\varphi(U(x_0)) = (\varphi_1 \cap \varphi_2)(U(x_0)) \subseteq G,$$

that is,

$$x \in U(x_0) \Longrightarrow (\varphi_1 \cap \varphi_2)(x) \subseteq G.$$

Therefore φ is upper semi-continuous (u.s.c.). $\square$

Semi-continuous Mappings



Semi-continuous Mappings

Corollary 1.1. If T is a compact space, a correspondence $\varphi : S \rightrightarrows T$ is closed-valued, then the correspondence φ is closed if and only if it is upper semi-continuous (u.s.c.).

Proof. If T is a compact space and φ is a closed-valued mapping, $\varphi(x)$ is a closed set for each $x \in S$, thus $\varphi(x)$ is a compact set and hence φ is compact-valued. If φ is u.s.c., then it is closed. Conversely, $\Delta : S \to \{T\}$ is the mapping such that $\Delta(x) = T$ for each x, Δ is a compact-valued u.s.c. correspondence. If φ is closed, then $\varphi = \varphi \cap \Delta$ is u.s.c. $\square$

Semi-continuous Mappings

Theorem 1.8. If S is a compact space and φ is a upper semi-continuous (u.s.c.) correspondence of S into itself such that, for each x, $\varphi(x) \neq \varnothing$, then there exists a non-empty compact subset K of S such that $\varphi(K) = K$.

Proof. The sets $S, \varphi(S), \varphi^2(S), \cdots$ form a sequence of non-empty compact sets; if any of these is equal to its successor, it is a set of the required kind and so the desired result is proved. We therefore assume that the sets are distinct; since $S \supseteq \varphi(S)$, we have $\varphi(S) \supseteq \varphi^2(S)$, $\varphi^2(S) \supseteq \varphi^3(S)$ and so on. Thus the sequence of sets $\varphi^n(S)$ is decreasing. By the finite intersection axiom, we have

$$K = \bigcap_{n=1}^{\infty} \varphi^n(S) \neq \varnothing.$$

For each n, we have $K \subseteq \varphi^{n-1}(S)$ and so $\varphi(K) \subseteq \varphi^n(S)$, whence $\varphi(K) \subseteq K$.

Semi-continuous Mappings

We now prove that $K \subseteq \varphi(K)$. Let a be a point of K; then, for each n, there exists a point x_n in $\varphi^n(S)$ such that $a \in \varphi(x_n)$. The sequence $\{x_n : n \in \mathbb{N}\}$ admits a cluster point x_0 and a sub-sequence $\{x_{k_n} : n \in \mathbb{N}\}$ converges to x_0. Since at most $n-1$ points of this sub-sequence are outside $\varphi^n(S)$ and $\varphi^n(S)$ is compact, we have $x_0 \in \varphi^n(S)$ and so $x_0 \in K$. Also $(x_{k_n}, a) \longrightarrow (x_0, a)$ and $(x_{k_n}, a) \in G(\varphi) = \sum_{x \in S} \varphi(x)$ and so, since φ is a closed mapping, $a \in \varphi(x_0)$; therefore $a \in \varphi(K)$. It follows that $K \subseteq \varphi(K)$ and hence that $\varphi(K) = K$. $\square$

Properties of Semi-continuous Mappings

In this section we compare the properties of lower semi-continuity with those of upper semi-continuity.

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^{-}(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap G \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \cap \varphi_2^{-}(G) \neq \varnothing\right\} \\
&= \varphi_1^{-}\left(\varphi_2^{-}(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^{-}(G)$ is open and so φ is lower semi-continuous (l.s.c.) $\square$

Properties of Semi-continuous Mappings

Theorem 2.1. If φ_1 is a lower semi-continuous (l.s.c.) correspondence of S into T and φ_2 is a lower semi-continuous (l.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a lower semi-continuous (l.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set in Z, then

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq F\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq F\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(F)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(F)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are lower semi-continuous (l.s.c.), $\varphi^+(F)$ is closed and so φ is lower semi-continuous (l.s.c.) $\square$

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 1. If G is an open set in Z, then

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \subseteq G\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \subseteq G\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^+(G)\right\} \\
&= \varphi_1^+\left(\varphi_2^+(G)\right).
\end{aligned}$$

Therefore, if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^+(G)$ is open and so φ is upper semi-continuous (u.s.c.) $\square$

Properties of Semi-continuous Mappings

Theorem 2.1′. If φ_1 is a upper semi-continuous (u.s.c.) correspondence of S into T and φ_2 is a upper semi-continuous (u.s.c.) correspondence of T into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is a upper semi-continuous (u.s.c.) correspondence of S into Z.

Proof 2. If F is a closed set of Z, then

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid (\varphi_2 \cdot \varphi_1)(x) \cap F \neq \varnothing\} \\
&= \{x \in S \mid \varphi_2(\varphi_1(x)) \cap F \neq \varnothing\} \\
&= \left\{x \in S \mid \varphi_1(x) \subseteq \varphi_2^{-}(F) \neq \varnothing\right\} \\
&= \varphi_1^{-}\left(\varphi_2^{-}(F)\right).
\end{aligned}$$

Therefore if φ_1 and φ_2 are upper semi-continuous (u.s.c.), $\varphi^{-}(F)$ is closed and so φ is upper semi-continuous (u.s.c.) $\square$

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} [\varphi_\lambda(x) \cap G] \neq \varnothing\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap G \neq \varnothing\} \\
&= \bigcup_{\lambda\in\Lambda} \varphi_\lambda^-(G).
\end{aligned}$$

Therefore the set $\varphi^-(G)$ is open and so φ is l.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.2. If every mapping φ_λ of S into T is a lower semi-continuous (l.s.c.) correspondence for $\lambda \in \Lambda$, then the union $\varphi = \bigcup_{\lambda\in\Lambda} \varphi_\lambda$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} \\
&= \left\{x \in S \mid \bigcup_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}F \subseteq \bigcap_{\lambda\in\Lambda} \mathbf{C}\left(\varphi_\lambda(x)\right)\right\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}F \subseteq \mathbf{C}\left(\varphi_\lambda(x)\right)\} \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq F\} \\
&= \bigcap_{\lambda\in\Lambda} \varphi_\lambda^+(F).
\end{aligned}$$

Therefore the set $\varphi^+(F)$ is closed and so φ is l.s.c. □

Properties of Semi-continuous Mappings

Theorem 2.2′**.** If every mapping φ_λ of S into T is a compact-valued u.s.c. correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a compact-valued u.s.c. correspondence of S into T.

Properties of Semi-continuous Mappings

Proof. By Theorem 1.5, the mapping φ_λ is compact-valued u.s.c., then the mapping $\varphi = \bigcap\limits_{\lambda\in\Lambda} \varphi_\lambda$ is closed. Suppose that $y_0 \notin \varphi(x_0) = \bigcap\limits_{\lambda\in\Lambda} \varphi_\lambda(x_0)$, then there exists $\lambda_0 \in$ such that $y_0 \notin \varphi_{\lambda_0}(x_0)$. There exist neighbourhood $U(x_0)$ and neighbourhood $V(y_0)$ such that

$$x \in U(x_0) \qquad \Longrightarrow \qquad \varphi_{\lambda_0}(x) \cap V(y_0) = \varnothing,$$

that is, $\varphi_{\lambda_0}(U(x_0)) \cap V(y_0) = \varnothing$, then $\bigcap\limits_{\lambda\in\Lambda} \varphi_\lambda(U(x_0)) \cap V(y_0) = \varnothing$ and hence $\varphi = \bigcap\limits_{\lambda\in\Lambda} \varphi_\lambda$ is closed.

The intersection $\varphi = \bigcap\limits_{\lambda\in\Lambda} \varphi_\lambda$ of a family of compact-valued mappings φ_λ is compact-valued from Non-empty Finite-intersection Theorem.

On the other hand, $\varphi = \bigcap\limits_{\lambda\in\Lambda} \varphi_\lambda = \varphi_{\gamma'} \cap \bigcap\limits_{\gamma\neq\gamma'} \varphi_\lambda$. The mapping $\varphi_{\gamma'}$ is compact-valued u.s.c., and the mapping $\bigcap\limits_{\gamma\neq\gamma'} \varphi_\lambda$ is closed, therefore, by Theorem 1.7, the mapping φ is u.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set of T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T\left(\varphi_\lambda(x)\right)\right\} \\
&= \bigcup_{\lambda\in\Lambda} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T\left(\varphi_\lambda(x)\right)\} = \bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq G\} = \bigcup_{\lambda\in\Lambda} \varphi_\lambda^+(G)
\end{aligned}$$

Therefore the set $\varphi^+(G)$ is open and so φ is u.s.c. □

Properties of Semi-continuous Mappings

Theorem 2.2′. If every mapping φ_λ of S into T is a upper semi-continuous (u.s.c.) correspondence for $\lambda \in \Lambda$, then the intersection $\varphi = \bigcap_{\lambda\in\Lambda} \varphi_\lambda$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set of T, we have

$$\begin{aligned}
\varphi^-(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcap_{\lambda\in\Lambda} \varphi_\lambda(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid F \subseteq \mathbf{C}_T \left(\bigcap_{\lambda\in\Lambda} \varphi_\lambda(x)\right)\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid F \subseteq \bigcup_{\lambda\in\Lambda} \mathbf{C}_T(\varphi_\lambda(x))\right\} = \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid F \subseteq \mathbf{C}_T(\varphi_\lambda(x))\}\right) \\
&= \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F = \varnothing\}\right) \\
&= \bigcap_{\lambda\in\Lambda} \{x \in S \mid \varphi_\lambda(x) \cap F \neq \varnothing\} = \bigcap_{\lambda\in\Lambda} \varphi_\lambda^-(F).
\end{aligned}$$

Therefore the set $\varphi^-(F)$ is closed and so φ is u.s.c. □

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$\begin{aligned}
\varphi^-(G) &= \{x \in S \mid \varphi(x) \cap G \neq \varnothing\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \cap G = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T G\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid G \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T(\varphi_n(x))\right\} = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid G \subseteq \mathbf{C}_T(\varphi_n(x))\}\right) \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T G\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G = \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap G \neq \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^-(G).
\end{aligned}$$

Therefore $\varphi^-(G)$ is an open set and so φ is l.s.c. $\square$

Properties of Semi-continuous Mappings

Pseudo-Theorem 2.3. If every mapping φ_n of S into T is a lower semi-continuous (l.s.c.) correspondence for $n = 1, \cdots, N$, then the intersection $\varphi = \bigcap_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^+(F) &= \{x \in S \mid \varphi(x) \subseteq F\} = \left\{x \in S \mid \bigcap_{n=1}^{N} \varphi_n(x) \subseteq F\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T \left(\bigcap_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T F \subseteq \bigcup_{n=1}^{N} \mathbf{C}_T\left(\varphi_n(x)\right)\right\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \mathbf{C}_T F \subseteq \mathbf{C}_T\left(\varphi_n(x)\right)\} \\
&= \bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq F\} = \bigcup_{n=1}^{N} \varphi_n^+(F).
\end{aligned}$$

Therefore $\varphi^+(F)$ is a closed set and so φ is l.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 1. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} = \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T \left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\varphi_n(x)\right)\} = \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a compacted-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a compacted-valued u.s.c. correspondence of S into T.

Proof 2. The set $\varphi(x) = \bigcup_{n=1}^{N} \varphi_n(x)$ is compact for all x. Moreover, if F is a closed set in T, we have

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(F).
\end{aligned}$$

Therefore $\varphi^{-}(F)$ is a closed set and so φ is u.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 1. If G is an open set in T, we have

$$\begin{aligned}
\varphi^+(G) &= \{x \in S \mid \varphi(x) \subseteq G\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq G\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\bigcup_{n=1}^{N} \varphi_n(x)\right)\right\} \\
&= \left\{x \in S \mid \mathbf{C}_T G \subseteq \bigcap_{n=1}^{N} \mathbf{C}_T \left(\varphi_n(x)\right)\right\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \mathbf{C}_T G \subseteq \mathbf{C}_T \left(\varphi_n(x)\right)\} \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq G\} = \bigcap_{n=1}^{N} \varphi_n^+(G).
\end{aligned}$$

Therefore $\varphi^+(G)$ is an open set and so φ is u.s.c. □

Properties of Semi-continuous Mappings

Theorem 2.3′. If every mapping φ_n of S into T is a upper semi-continuous (u.s.c.) correspondence for $n = 1, \cdots, N$, then the union $\varphi = \bigcup_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into T.

Proof 2. If F is a closed set in T, we have

$$\begin{aligned}
\varphi^{-}(F) &= \{x \in S \mid \varphi(x) \cap F \neq \varnothing\} = \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F \neq \varnothing\right\} \\
&= \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \cap F = \varnothing\right\} = \mathbf{C}_S \left\{x \in S \mid \bigcup_{n=1}^{N} \varphi_n(x) \subseteq \mathbf{C}_T F\right\} \\
&= \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \subseteq \mathbf{C}_T F\}\right) = \mathbf{C}_S \left(\bigcup_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F \neq \varnothing\}\right) \\
&= \bigcap_{n=1}^{N} \{x \in S \mid \varphi_n(x) \cap F = \varnothing\} = \bigcap_{n=1}^{N} \varphi_n^{-}(F).
\end{aligned}$$

Therefore $\varphi^{-}(F)$ is a closed set and so φ is u.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned}
\varphi^- \left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} G_n^k \neq \varnothing\right\} \\
&= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap G_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap G_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^- \left(G_n^k\right).
\end{aligned}$$

Then $\varphi^- \left(E^k\right)$ is an open set and therefore so is $\varphi^-(G) = \bigcup_k \varphi^- \left(E^k\right)$.

$$\begin{aligned}
\varphi^-(G) &= \varphi^- \left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcup_k E^k\right) \neq \varnothing\right\} \\
&= \left\{x \in S \mid \bigcup_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\
&= \bigcup_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcup_k \varphi^- \left(E^k\right).
\end{aligned}$$

Hence φ is l.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.4. If every mapping φ_n is a lower semi-continuous (l.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a lower semi-continuous (l.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} F_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq F_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(F_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is a closed set and therefore so is $\varphi^+(F) = \bigcap_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(F) &= \varphi^+\left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcap_k E^k\right\} \\ &= \bigcap_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcap_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is l.s.c. □

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} Y_n$.

Proof 1. Let $G = \bigcup_k E^k$ be the union of the elementary open sets E^k in T. Since $E^k = \prod_{n=1}^{N} G_n^k$, where G_n^k is an open set in T_n, we have

$$\begin{aligned} \varphi^+\left(E^k\right) &= \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \subseteq \prod_{n=1}^{N} G_n^k\right\} \\ &= \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \subseteq G_n^k\right\} = \bigcap_{n=1}^{N} \varphi_n^+\left(G_n^k\right). \end{aligned}$$

Then $\varphi^+\left(E^k\right)$ is an open set and therefore so is $\varphi^+(G) = \bigcup_k \varphi^+\left(E^k\right)$.

$$\begin{aligned} \varphi^+(G) &= \varphi^+\left(\bigcup_k E^k\right) = \left\{x \in S \mid \varphi(x) \subseteq \bigcup_k E^k\right\} \\ &= \bigcup_k \left\{x \in S \mid \varphi(x) \subseteq E^k\right\} = \bigcup_k \varphi^+\left(E^k\right). \end{aligned}$$

Hence φ is u.s.c. □

Properties of Semi-continuous Mappings

Theorem 2.4′. If every mapping φ_n is a upper semi-continuous (u.s.c.) correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a upper semi-continuous (u.s.c.) correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof 2. Let $F = \bigcap_k E^k$ be the intersection of the elementary closed sets E^k in T. Since $E^k = \prod_{n=1}^{N} F_n^k$, where F_n^k is a closed set in T_n, we have

$$\begin{aligned}
\varphi^- \left(E^k\right) &= \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \left\{x \in S \mid \prod_{n=1}^{N} \varphi_n(x) \cap \prod_{n=1}^{N} F_n^k \neq \varnothing\right\} \\
&= \left\{x \in S \mid \prod_{n=1}^{N} \left(\varphi_n(x) \cap F_n^k\right) \neq \varnothing\right\} = \bigcap_{n=1}^{N} \left\{x \in S \mid \varphi_n(x) \cap F_n^k \neq \varnothing\right\} = \bigcap_{n=1}^{N} \varphi_n^- \left(F_n^k\right).
\end{aligned}$$

Then $\varphi^- \left(E^k\right)$ is a closed set and therefore so is $\varphi^-(F) = \bigcap_k \varphi^- \left(E^k\right)$.

$$\begin{aligned}
\varphi^-(F) &= \varphi^- \left(\bigcap_k E^k\right) = \left\{x \in S \mid \varphi(x) \cap \left(\bigcap_k E^k\right) \neq \varnothing\right\} \\
&= \left\{x \in S \mid \bigcap_k \left[\varphi(x) \cap E^k\right] \neq \varnothing\right\} \\
&= \bigcap_k \left\{x \in S \mid \varphi(x) \cap E^k \neq \varnothing\right\} = \bigcap_k \varphi^- \left(E^k\right).
\end{aligned}$$

Hence φ is u.s.c. □

Properties of Semi-continuous Mappings

Theorem 2.4′**.** If every mapping φ_n is a compact-valued u.s.c. correspondence of S into T_n for $n = 1, \cdots, N$, then the Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ is also a compact-valued u.s.c. correspondence of S into $T = \prod_{n=1}^{N} T_n$.

Proof. To fix our ideas, we consider the case $N = 2$. Since $\varphi_1(x)$ and $\varphi_2(x)$ are compact, so is $\varphi(x) = \varphi_1(x) \times \varphi_2(x)$, by Tychonoff's Theorem.

Properties of Semi-continuous Mappings

Let G be an open set in T and. let a be an element of $\varphi^+(G)$. Since $\varphi(a)$ is compact and is contained in G, it can be covered by a finite number of elementary open sets contained in G; let $E^1, \cdots, E^K$ be such a finite covering. Suppose that $(y_1, y_2) \in \varphi_1(a) \times \varphi_2(a)$; let $E(y_1)$ denote the union of the E^k meeting the set $\{y_1\} \times \varphi_2(a)$ and let $E(y_2)$ be the union of the E^k meeting the set $\varphi_1(a) \times \{y_2\}$. The projection $\pi_2 E(y_1)$ of the set $E(y_1)$ on the space Y_2 is open, by Theorem 2 of §9, Chapter IV. As y_1 varies the sets. $\pi_2 E(y_1)$ are finite in number and therefore their intersection is open; similarly the intersection of the sets $\pi_1 E(y_2)$ is open. We write

$$G_1 = \bigcap_{y_2 \in \varphi_2(a)} \pi_1 E(y_2) \qquad \text{and} \qquad G_2 = \bigcap_{y_1 \in \varphi_1(a)} \pi_2 E(y_1).$$

Then $E = G_1 \times G_2$ is an elementary open set of T and

$$\varphi^+(E) = \{x \mid \varphi_1(x) \times \varphi_2(x) \subseteq G_1 \times G_2\} = \left[\varphi_1^+(G_1) \cap \varphi_2^+(G_2)\right] \cup \varphi_1^+(\varnothing) \cup \varphi_2^+(\varnothing).$$

Therefore $\varphi^+(E)$ is an open set of S; since $\varphi(a) \subseteq E \subseteq G$, we also have

$$a \in \varphi^+(E) \subseteq \varphi^+(G)$$

Therefore the set $\varphi^+(G)$ is a neighbourhood of each of its points and so is open; hence φ is compact-valued u.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.5′. Let S and T be topological vector spaces. If every mapping $\varphi_n : X \to Y$ is a compact-valued u.s.c. correspondence for $n = 1, \cdots, N$, then the sum $\varphi = \sum_{n=1}^{N} \varphi_n$ is a compact-valued u.s.c. correspondence.

Theorem 2.5′. The sum $\varphi = \sum_{n=1}^{N} \varphi_n$ of a finite number of compact-valued u.s.c. correspondences φ_n of S into T is a compact-valued u.s.c. correspondence of S into T.

Properties of Semi-continuous Mappings

Proof. We use mathematical induction to verify this theorem.

(1) If $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $U(x_0)$ such that $x \in U(x_0)$ implies $\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq G$.

(1) Let T be a topological vector space. Define $f : T^N \to T$ by $f(y_1, \cdots, y_N) = \sum_{n=1}^{N} y_n$, then f is a continuous function, $f(\varphi_1 x_0, \cdots, \varphi_n(x_0)) = \sum_{n=1}^{N} \varphi_n(x_0)$. As you know, $\varphi_n(x_0)$ is compact for $n = 1, \cdots, N$, then $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0)$ is compact.

Properties of Semi-continuous Mappings

(2) For any open set $G \subseteq T$ with $\varphi(x_0) = \sum_{n=1}^{N} \varphi_n(x_0) \subseteq G$, there exists a neighborhood $V = U(0)$ of origin such that $\sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$, there exist N neighborhoods $V_1, \cdots, V_N$ with $\sum_{n=1}^{N} V_n \subseteq V$ such that $\sum_{n=1}^{N} [\varphi_n(x_0) + V_n] = \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G$. $\varphi_n : X \to Y$ is u.s.c. for $n = 1, \cdots, N$, then $\varphi_n(x_0) \subseteq \varphi_n(x_0) + V_n$, there exists a neighborhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad x \in \varphi_n(x_0) + V_n.$$

For any $x \in N(x_0) \equiv \bigcap_{n=1}^{N} U_n(x_0)$,

$$\varphi(x) = \sum_{n=1}^{N} \varphi_n(x) \subseteq \sum_{n=1}^{N} [\varphi_n(x_0) + V_n] \subseteq \sum_{n=1}^{N} \varphi_n(x_0) + V \subseteq G.$$

Thus $\varphi = \sum_{n=1}^{N} \varphi_n$ is compact-valued and u.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.6. Let T be a linear vector space. If $\varphi : S \to \mathbb{R}^N$ is a lower semi-continuous (l.s.c.) correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_0 \in \mathbf{CE}(\varphi)(x_0) = \mathbf{CE}(\varphi(x_0))$,

$$y_0 = \sum_{k=1}^{K} \lambda^k z_0^k, \qquad z_0^k \in \varphi(x_0), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a sequence $z_n^k \to z_0^k$ such that $z_n^k \in \varphi(x_n)$, $y_n = \sum_{k=1}^{K} \lambda^k z_n^k \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_n))$. Therefore $\mathbf{CE}(\varphi)$ is lower semi-continuous (l.s.c.). $\square$

Properties of Semi-continuous Mappings

Theorem 2.6′. If $\varphi : S \to \mathbb{R}^N$ is a compact-valued u.s.c. correspondence, then its convex envelope $\mathbf{CE}(\varphi)$ is a compact-valued u.s.c. correspondence.

Proof. According to Caratheodory Characterization Theorem, for any $y_n \in \mathbf{CE}(\varphi)(x_n) = \mathbf{CE}(\varphi(x_n))$,

$$y_n = \sum_{k=1}^{K} \lambda_n^k z_n^k, \qquad z_n^k \in \varphi(x_n), \quad \sum_{k=1}^{K} \lambda_n^k = 1, \quad \lambda_n^k \geqslant 0.$$

For any $x_n \to x_0$, there exists a convergent subsequence $\{z_{n'}^k, n' \in \mathbb{N}\} \subseteq \{z_n^k, n \in \mathbb{N}\}$ such that $\lim_{n'} z_{n'}^k \to z_0^k \in \varphi(x_0)$, then there corresponds a subsequence $\{y_{n'}, n' \in \mathbb{N}\} \subseteq \{y_n, n \in \mathbb{N}\}$ such that $y_{n'} = \sum_{k=1}^{K} \lambda_{n'}^k z_{n'}^k \to \sum_{k=1}^{K} \lambda_n^k z_0^k = y_0 \in \varphi(x_0) \subseteq \mathbf{CE}(\varphi(x_0))$.
Therefore $\mathbf{CE}(\varphi)$ is compact-valued and u.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.7. $\varphi : S \to T$ is a lower semi-continuous (l.s.c.) correspondence if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a lower semi-continuous (l.s.c.) correspondence.

Proof. For any open set $G \subseteq T$ with $\varphi(x_0) \subseteq G$ equivalently $\overline{\varphi}(x_0) \subseteq G$, there exists a neighbourhood $U(x_0)$ such that, for any $x \in U(x_0)$, $\varphi(x) \subseteq G$ equivalently $\overline{\varphi}(x) \subseteq G$. Thus φ is l.s.c. if and only if its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is l.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 2.7′**.** Let T be a regular metric space. If $\varphi : S \to T$ is a u.s.c. correspondence, then its convex hull $\mathbf{CH}(\varphi) = \overline{\varphi}$ is a u.s.c. correspondence.

Proof. For any open set $G \subseteq T$ with $\overline{\varphi}(x_0) \subseteq G$, $\overline{\varphi}(x_0) \cap (\mathbf{C}_T G) = \varnothing$. According to the regularity of metric space, there exists an open set G' and O such that

$$\overline{\varphi}(x_0) \subseteq G', \qquad \mathbf{C}_T G \subseteq O, \qquad G' \cap O = \varnothing$$

As φ is u.s.c., there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad \varphi(x) \subseteq G',$$

that is, $\varphi(U(x_0)) \subseteq G'$. Since $G' \cap (\mathbf{C}_T G) = \varnothing$, then $\overline{\varphi}(U(x_0)) \subseteq G$. Therefore $\overline{\varphi}$ is u.s.c. $\square$

Semi-continuous Correspondences

Definition 3.1 Let X and Y be metric spaces, and let $\varphi : X \rightrightarrows Y$ be a correspondence.

(1) We say that φ is inner semi-continuous (i.s.c.) if whenever $x_n \to x$ and $y \in \varphi(x)$, there is a sequence $y_n \in \varphi(x_n)$ for each n such that $y_n \to y$.

(2) We say that φ is outer semi-continuous (o.s.c.) if whenever $x_n \to x$ and $y_n \in \varphi(x_n)$ for each n, there is a convergent subsequence $y_{n_k} \to y$ with $y \in \varphi(x)$.

Semi-continuous Correspondences

Theorem 3.1 (Lower Semi-continuity and Inner Semi-continuity) Let X and Y be two metric spaces. A correspondence $\varphi : X \rightrightarrows Y$. φ is l.s.c. at x if and only if it is i.s.c. at x.

Proof. Suppose φ is not l.s.c. at some $x_0 \in X$. Then there exists an open subset G of Y such that $\varphi(x_0) \cap G \neq \varnothing$ and, for every $n \in \mathbb{N}$, there exists an $x_n \in B_{\frac{1}{n}}(x_0)$ with $\varphi(x_n) \cap G = \varnothing$. Note that $x_n \to x_0$, and pick any $y_0 \in \varphi(x_0) \cap G$. By hypothesis, there must exist a sequence $\{y_n : n \in \mathbb{N}\} \subseteq Y$ such that $y_n \to y_0$ and $y_n \in \varphi(x_n)$ for each n. But since $y_0 \in G$ and G is open, $y_n \in G$ for n large enough, contradicting that $\varphi(x_n) \cap G = \varnothing$ for all n.

Semi-continuous Correspondences

Conversely, if $x_n \to x_0$ and $y_0 \in \varphi(x_0)$, then $y_0 \in \varphi(x_0) \cap B_{\frac{1}{n}}(y_0)$. By lower semi-continuity, there exists a $\varepsilon_n > 0$ with $\lim_{n\to\infty} \varepsilon_n = 0$ such that

$$x \in B_{\varepsilon_n}(x_0) \quad \Longrightarrow \quad \varphi(x) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing.$$

Since $x_n \to x_0$, so there exists k_n such that for all $n \geqslant k_n$, $x_n \in B_{\varepsilon_n}(x_0)$, then $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$. We may assume k_n is increasing.

For $n < k_n$, we pick $y_n \in \varphi(x_n)$. For any $k_n \leqslant n < k_{n+1}$, we have $\varphi(x_n) \cap B_{\frac{1}{n}}(y_0) \neq \varnothing$, pick $y_n \in \varphi(x_n) \cap B_{\frac{1}{n}}(y_0)$. Then $d(y_n, y_0) < \dfrac{1}{n}$. $y_n \to y_0$. Hence φ is i.s.c. at x_0. $\square$

Semi-continuous Correspondences

Theorem 3.1′ (Upper Semi-continuity and outer Semi-continuity) Let X and Y be two metric spaces. Mapping $\varphi : X \rightrightarrows Y$ is a correspondence.

(1) If φ is compact-valued u.s.c. at x, then it is o.s.c. at x.

(2) If φ is o.s.c. at x, then φ is u.s.c. at x.

Proof of (1). Suppose φ is u.s.c. at x_0, $x_n \to x_0$ and $y_n \in \varphi(x_n)$. Since φ is compact-valued, $\varphi(x_0)$ has a bounded neighborhood U. Then $S = \{x_0, x_1, x_2, \cdots\}$ is sequentially compact in X, hence S is compact in X. By Proposition 1, $\varphi(S)$ is sequentially compact in Y. So, if $y_n \in \varphi(x_n)$ for each n, then $\{y_n : n \in \mathbb{N}\}$ — being a sequence in $\varphi(S)$ – must possess a subsequence that converges to some $y_0 \in \varphi(S)$. With an innocent abuse of notation, we denote this subsequence also by $\{y_n : n \in \mathbb{N}\}$, and write $y_n \to y_0$.

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$, and observe that this implies $\varepsilon = d_Y(y_0, \varphi(x_0)) > 0$ because $\varphi(x_0)$ is a compact (hence closed) set. Now define

$$T = \left\{ z \in Y : d_Y(z, \varphi(x_0)) \leqslant \frac{\varepsilon}{2} \right\}$$

Since $d_Y(z, \varphi(x_0))$ is a continuous map on z, T is a closed set. Moreover, by definition, we have

$$\varphi(x_0) \subseteq T^\circ$$

and $y_0 \notin T$. But, since φ is u.s.c. at x_0, there exists $\delta > 0$ such that

$$\varphi(B_\delta(x_0)) \subseteq T^\circ.$$

Thus there is $n \in \mathbb{N}$ such that $y_n \in \varphi(x_n) \subseteq T^\circ$ for all $n \geqslant N$. But then, since T is a closed set and $y_n \to y_0$, we must have $y_0 \in T$, a contradiction.

Semi-continuous Correspondences

We wish to show that $y_0 \in \varphi(x_0)$. Let $y_0 \notin \varphi(x_0)$. Note that $\varphi(x_0)$ is compact, there exist two neighbourhoods $V(y_0)$ and $V'(\varphi(x_0))$ such that $V(y_0) \cap V'(\varphi(x_0)) = \varnothing$, then $y_0 \notin \overline{V'}(\varphi(x_0))$.

If φ is u.s.c. at x_0 and $x_n \to x_0$, then there exists a $N \in \mathbb{N}$ such that $\varphi(x_n) \subseteq V'(\varphi(x_0))$ for $n \geqslant N$, thus $y_n \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, and $y_0 \in \overline{V'}(\varphi(x_0))$ for $n \geqslant N$, which contradicts $y_0 \notin \overline{V'}(\varphi(x_0))$.

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then there exists an open subset G of Y with $\varphi(x_0) \subseteq G$ and

$$\varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G \neq \varnothing$$

for any $n \in \mathbb{N}$. But then there exist $(x_n) \subseteq X$ and $(y_n) \subseteq Y$ such that $x_n \to x_0$ and $y_n \in \varphi\left(B_{\frac{1}{n}}(x_0)\right) \setminus G$ for each n. Since $\varphi(x_0) \subseteq G$ and each y_n belongs to the closed set $Y \setminus G$, no subsequence of (y_n) can converge to a point in $\varphi(x_0)$. $\square$

Semi-continuous Correspondences

Proof of (2). Now suppose that for every sequence $x_n \to x_0$ and $y_n \in \varphi(x_n)$, there is a subsequence of y_n with limit in $\varphi(x)$.

Suppose φ is not u.s.c., then for any open set $G \subseteq Y$ with $\varphi(x_0) \subseteq G$ and any neighbourhood $U_n(x_0)$, there exist $x_n \in U_n(x_0)$ such that $\varphi(x_n) \nsubseteq G$. As $x_n \to x_0$, there exists $y_n \in \varphi(x_n)$ and $y_n \notin G$. Then there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $y_{n'} \to \varphi(x_0)$. However, $y_n \notin G$, there exists a convergent subsequence $\{y_{n'}; n' \in \mathbb{N}\} \subseteq \{y_n; n \in \mathbb{N}\}$ such that $\lim_{n' \to \infty} y_{n'} \notin G$, then $\lim_{n' \to \infty} y_{n'} \notin \varphi(x_0)$. This is a contradiction! $\square$

Properties of Semi-continuous Mappings

Theorem 3.2. (Inner Semi-continuity) If φ_1 is an i.s.c. mapping of X into Y and φ_2 is an i.s.c. mapping of Y into Z, then the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an i.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_0 \in \varphi(x_0) = (\varphi_2 \cdot \varphi_1)(x_0) = \varphi_2(\varphi_1(x_0))$, then there exists $y_0 \in \varphi_1(x_0)$ such that $z_0 \in \varphi_2(y_0)$.

Whenever $x_n \to x_0$ and $y_0 \in \varphi_1(x_0)$, by inner semi-continuity of φ_1, there exists a sequence $y_n \in \varphi_1(x_n)$ such that $y_n \to y_0$. On the other hand, $z_0 \in \varphi_2 y_0$, by inner semi-continuity of φ_2, there exists a sequence $z_n \in \varphi_2(y_n)$ such that $z_n \to z_0$.

$[y_n \in \varphi_1(x_n)$ and $z_n \in \varphi_2(y_n)]$ imply $z_n \in \varphi_2(\varphi_1(x_n)) = \varphi(x_n)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is i.s.c. $\square$

Properties of Semi-continuous Mappings

Theorem 3.2′ (Outer Semi-continuity) If φ_1 is an o.s.c. mapping of X into Y and φ_2 is an o.s.c. mapping of Y into Z, the composition product $\varphi = \varphi_2 \cdot \varphi_1$ is an o.s.c. mapping of X into Z.

Proof. If $x_n \to x_0$ and $z_n \in \varphi(x_n) = (\varphi_2 \cdot \varphi_1)(x_n) = \varphi_2(\varphi_1(x_n))$, then there exists $y_n \in \varphi_1(x_n)$ such that $z_n \in \varphi_2(y_n)$.

Whenever $x_n \to x_0$ and $y_n \in \varphi_1(x_n)$, by outer semi-continuity of φ_1, there exists a convergent subsequence $y_{n_1} \to y_0$ with $y_0 \in \varphi_1(x_0)$. On the other hand, $z_n \in \varphi_2(y_n)$, then $z_{n_1} \in \varphi_2(y_{n_1})$. By outer semi-continuity of φ_2, there exists a convergent subsequence $z_{n_2} \to z_0$ with $z_0 \in \varphi_2(y_0)$.

$[y_0 \in \varphi_1(x_0)$ and $z_0 \in \varphi_2(y_0)]$ imply $z_0 \in \varphi_2(\varphi_1(x_0)) = \varphi(x_0)$, thus $\varphi = \varphi_2 \cdot \varphi_1$ is o.s.c. $\square$

Semi-continuous Correspondences

Theorem 3.3 (Inner Semi-continuity) The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of i.s.c. mappings φ_n of X into Y_n is an i.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $\left(y_1^0, \cdots, y_N^0\right) = y^0 \in \varphi\left(x^0\right) = \prod_{n=1}^{N} \varphi_n\left(x^0\right)$, then $y_n^0 \in \varphi_n\left(x^0\right)$ for $n = 1, \cdots, N$. If every φ_n is i.s.c., then there is a sequence $y_n^q \in \varphi_n\left(x^q\right)$ for each q such that $y_n^q \to y_n^0$. Thus $y^q = \left(y_1^q, \cdots, y_N^q\right) \in \varphi\left(x^q\right) = \prod_{n=1}^{N} \varphi_n\left(x^q\right)$ such that $y^q = \left(y_1^q, \cdots, y_N^q\right) \to \left(y_1^0, \cdots, y_N^0\right) = y^0$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is i.s.c. $\square$

Semi-continuous Correspondences

Theorem 3.3′ **(Outer Semi-continuity)** The Cartesian product $\varphi = \prod_{n=1}^{N} \varphi_n$ of a finite number of o.s.c. mappings φ_n of X into Y_n is an o.s.c. mapping of X into $Y = \prod_{n=1}^{N} Y_n$.

Proof. If $x^q \to x^0$ and $(y_1^q, \cdots, y_N^q) = y^q \in \varphi(x^q) = \prod_{n=1}^{N} \varphi_n(x^q)$, then $y_n^q \in \varphi_n(x^q)$ for $n = 1, \cdots, N$. If φ_1 is o.s.c., then there is a convergent subsequence $y_1^{q_1} \to y_1^0$ with $y_1^0 \in \varphi_1 x^0$. In this case, $y_n^{q_1} \in \varphi_n x^{q_1}$ for $n = 1, \cdots, N$. If φ_2 is o.s.c., then there is a convergent subsequence $y_n^{q_2} \to y_n^0$ with $y_n^0 \in \varphi_n(x^0)$ for $n = 1, 2$. In this case, $y_n^{q_2} \in \varphi_n x^{q_2}$ for $n = 1, \cdots, N$. Follow these steps, we can obtain that, if φ_N is o.s.c., then there is a convergent subsequence $y_n^{q_N} \to y_n^0$ with $y_n^0 \in \varphi_n(x^0)$ for $n = 1, \cdots, N$. Hence $y^{q_N} = (y_1^{q_N}, \cdots, y_N^{q_N}) \to (y_1^0, \cdots, y_N^0) = y^0$ and $y^0 = (y_1^0, \cdots, y_N^0) \in \prod_{n=1}^{N} \varphi_n(x^0) = \varphi(x^0)$. Therefore $\varphi = \prod_{n=1}^{N} \varphi_n$ is o.s.c.

□

Berge's Maximum Theorem

Theorem 4.1. If f is a l.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \sup\{f(x, y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. Suppose that $x_0 \in S$ and let y_0 be such that

$$y_0 \in \varphi(x_0); \qquad f(x_0, y_0) \geqq M(x_0) - \varepsilon.$$

There exist neighbourhoods $U(x_0)$ and $V(y_0)$ such that

$$(x, y) \in U(x_0) \times V(y_0) \Longrightarrow f(x, y) \geqq f(x_0, y_0) - \varepsilon \geqq M(x_0) - 2\varepsilon;$$

there exists a neighbourhood $U'(x_0)$ such that

$$x \in U'(x_0) \Longrightarrow \varphi(x) \cap V(y_0) \neq \varnothing.$$

Therefore

$$x \in U(x_0) \cap U'(x_0) \Longrightarrow M(x) \geqq M(x_0) - 2\varepsilon.$$

□

Berge's Maximum Theorem

Theorem 4.2. If f is a u.s.c. numerical function in $S \times T$ and φ is a compact-valued u.s.c. correspondence of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the mumerical function M defined by

$$M(x) = \max\{f(x, y) \mid y \in \varphi(x)\}$$

is u.s.c.

Berge's Maximum Theorem

Proof. Suppose that $x_0 \in S$; to each y in $\varphi(x_0)$ there correspond neighbourhoods $U_y(x_0)$ and $V(y)$ such that

$$(x,z) \in U_y(x_0) \times V(y) \implies f(x,z) \leqq f(x_0,y) + \varepsilon.$$

Since $\varphi(x_0)$ is compact, it can be covered by a finite number of neighbourhoods of the form $V(y)$, say $V(y_1), V(y_2), \cdots, V(y_N)$.

Putting $U'(x_0) = \bigcap_{n=1}^{N} U_{y_n}(x_0)$ and $V(\varphi(x_0)) = \bigcup_{n=1}^{N} V(y_n)$, we have

$$x \in U'(x_0), y \in V(\varphi(x_0)) \implies f(x,y) \leqq \max_n f(x_0,y_n) + \varepsilon \leqq M(x_0) + \varepsilon.$$

Moreover there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \implies \varphi(x) \subseteq V(\varphi(x_0))$$

and so

$$x \in U(x_0) \cap U'(x_0) \implies M(x) = \max_{y \in \varphi(x)} f(x,y) \leqq M(x_0) + \varepsilon.$$

□

Berge's Maximum Theorem

Theorem 4.3 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \max\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $f(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $f = \varphi \cap \Delta$ is u.s.c. $\square$

Berge's Maximum Theorem

Let f be a continuous numerical function defined in a topological space T. A family of compact sets $\mathscr{K} = \{K_i \mid i \in I\}$ in T is called selective (with respect to f) if for each i there exists one and only one a_i such that

$$a_i \in K_i; \qquad f(a_i) = \max\{f(y) \mid y \in K_i\}.$$

In other words, the maximum of f is attained at only one point of the set K_i. For example, in $\mathbb{R}^N$ every family of balls is selective, with respect to $f(y) = \pi_i y$; in $\mathbb{R}$, every family of compact sets is selective with respect to $f(y) = y$.

Berge's Maximum Theorem

Theorem 4.4. Let φ be a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$. If the family of sets $\{\varphi(x) \mid x \in S\}$ is selective, there is a single-valued continuous mapping σ of S into T such that, for each x, $\sigma(x) \in \varphi(x)$.

Proof. Let f be a continuous numerical function in T for which $\{\varphi(x) \mid x \in S\}$ is selective; if $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ then ϕ is a single-valued mapping of S into T. Moreover, it is u.s.c. by the preceding theorem and so is continuous (since it is single-valued). The mapping $\sigma(x) = \phi(x)$ satisfies

$$\sigma(x) \in \varphi(x) \quad \text{for all} \quad x \in S$$

as required. $\square$

Berge's Maximum Theorem

Corollary 4.1. If φ is a continuous mapping of X into $\mathbb{R}$ such that, for each x, $\varphi(x) \neq \varnothing$, there exists a continuous single-valued mapping σ such that, for each x, $\sigma(x) \in \varphi(x)$

Proof. It is sufficient to take $f(y) = y$, whence $\sigma(x) = \max \varphi(x)$. □

Berge's Maximum Theorem

Theorem 4.5. If f is a l.s.c. numerical function in $S \times T$ and φ is a compact-valued u.s.c. correspondence of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is l.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x_0) - \varepsilon \leqslant M(x).$$

Berge's Maximum Theorem

If φ is compact-valued, then there exist $y_1, \cdots, y_K \in \varphi(x_0)$ such that $\varphi(x_0) \subseteq \bigcup_{k=1}^{K} V_k(y_k)$. If φ is u.s.c., then there exists a neighbourhood $U_0(x_0)$ such that

$$x \in U_0(x_0) \Longrightarrow \varphi(x) \subseteq \bigcup_{k=1}^{K} V_k(y_k).$$

If f is l.s.c., then, for any $\varepsilon > 0$, $y_k \in \varphi(x_0)$, there exists a neighbourhood $U_{y_k}(x_0)$ such that, for any $x \in U_{y_k}(x_0)$ and any $y \in V_k(y_k)$, $f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Define $U(x_0) = U_0(x_0) \cap \bigcap_{k=1}^{K} U_{y_k}(x_0)$. Note that $y_1, \cdots, y_K \in \varphi(x_0)$, then, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y_k \in \varphi(x_0)} f(x_0, y_k) - \varepsilon \leqslant f(x, y)$.

Therefore, for any $x \in U(x_0)$ and $y \in \varphi(x)$,
$M(x_0) - \varepsilon \leqslant \inf_{y \in \varphi(x)} f(x, y) = M(x_0)$. $\square$

Berge's Maximum Theorem

Theorem 4.6. If f is a u.s.c. numerical function in $S \times T$ and φ is a l.s.c. mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \inf\{f(x, y) \mid y \in \varphi(x)\}$$

is u.s.c.

Proof. We want to prove that, for any $\varepsilon > 0$, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \Longrightarrow M(x) \leqslant M(x_0) + \varepsilon.$$

Berge's Maximum Theorem

From the definition of infimum, for any $\varepsilon > 0$, there exists $y_0 \in \varphi(x_0)$ such that $f(x_0, y_0) \leqslant M(x_0) + \frac{\varepsilon}{2}$.

If f is u.s.c., then there exist two neighbourhoods $U_1(x_0)$ and $V(y_0)$ such that, for any $x \in U_1(x_0)$ and any $y \in V(y_0)$, $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2}$.

Therefore $f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon$.

If φ is l.s.c., for $\varphi(x_0) \cap V(y_0) \neq \varnothing$, then there exists a neighbourhood $U_2(x_0)$ such that

$$x \in U_2(x_0) \Longrightarrow \varphi(x) \subseteq \varphi(x) \cap V(y_0) \neq \varnothing.$$

Define $U(x_0) = U_1(x_0) \cap U_2(x_0)$. for any $x \in U(x_0)$, there exists $y \in V(y_0)$, such that

$$M(x) = \inf_{y \in \varphi(x)} f(x, y) \leqslant f(x, y) \leqslant f(x_0, y_0) + \frac{\varepsilon}{2} \leqslant M(x_0) + \varepsilon.$$

Therefore M is u.s.c. $\square$

Berge's Maximum Theorem

Theorem 4.7 (Maximum Theorem). If f is a continuous numerical function in T and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, then the numerical function M defined by $M(x) = \min\{f(y) \mid y \in \varphi(x)\}$ is continuous in X and the mapping f defined by $\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$ is a u.s.c. correspondence of S into T.

Proof. The function f is continuous in $S \times T$ and so M is a continuous numerical function; moreover the mapping Δ given by

$$\Delta x = \{y \in T \mid M(x) - f(y) \leqq 0\}$$

is closed and hence $\phi = \varphi \cap \Delta$ is u.s.c. $\square$

Berge's Maximum Theorem

In economic activities, decision makers often change their preferences according to the situation of the environment, so the preference in economic activities is often changed by the environment on which preferences depend. Suppose $\beta : S \to T$ is a set-valued mapping, for any $x \in S$, $\prec_x$ is a binary relationship defined on $\beta(x)$ (set-valued mapping), which is transitive, but is not reflexive and complete. Set $\varphi(x)$ to be the set of maximal elements in $\beta(x)$ according to $\prec_x$.

Berge's Maximum Theorem

Theorem 4.8. Suppose that $\beta : S \to T$ is a correspondence, $\varphi(x) = \left\{ y \in T \mid y \in \max_{\prec_x} \beta(x) \right\}$, and $\{(x, y, z) \in S \times T \times T \mid y \in \beta(x), z \in \beta(x), y \not\prec_x z\}$ is closed. If $\beta(x)$ is compact, then $\varphi(x) \neq \varnothing$ is compact, β is continuous, φ is u.s.c.

Berge's Maximum Theorem

Proof. If $\varphi(x) = \varnothing$, then for any $y \in \beta(x)$, there exists $y' \in \beta(x)$ such that $y \prec_x y'$. Define $r(z) = \{y \in \beta(x) \mid y \prec_x z\}$, then $\beta(x) \subseteq \bigcup_{z \in \beta(z)} \{r(z)\}$. Note that $\beta(x)$ is compact and $\bigcup_{z \in \beta(z)} \{r(z)\}$ is its open covering, then there exist $r(z_1), \cdots, r(z_K)$, such that $\beta(x) \subseteq \bigcup_{k=1}^{K} r(z_k)$. $\{z_1, \cdots, z_K\}$ in finite, then there is the maximal to be z_K, as $\prec_x$ is transitive, then for $k = 1, \cdots, K-1$, $z_K \notin r(z_k)$. But $\prec_x$ is not reflexive, then $z_K \notin r(z_K)$, and hence $z_K \notin \beta(x)$, which is a contradiction. Therefore $\varphi(x) \neq \varnothing$.

Berge's Maximum Theorem

According to the continuity of $\prec_x$, $\{y \in \beta(x) \mid y \not\prec_x z\}$ is closed, then

$$\varphi(x) = \{y \in T \mid z \in \beta(x), y \not\prec_x z\} = \bigcap_{z \in \beta(x)} \{y \in \beta(x) \mid y \not\prec_x z\}$$

is closed, $\varphi(x) \neq \varnothing$ and $\varphi(x) \subseteq \beta(x)$, $\beta(x)$ is compact, then $\varphi(x)$ is compact.

In order to check that φ is u.s.c.at x_0, we want to prove that φ is closed-valued,

$$\text{for any } (x_n, y_n) \to (x, y),\ y_n \in \varphi(x_n) \quad \Longrightarrow \quad y \in \varphi(x).$$

Since β is l.s.c., then, for $z \in \beta(x)$, there exists $z_n \to z$ such that, for any n, $z_n \in \varphi(x_n)$, then $y_n \not\prec_{x_n} z_n$. From the continuity of $\prec_x$, $x_n \to x$, $y_n \to y$, $z_n \to z$, then $y \not\prec_x z$, and hence $y = \max_{\prec_x} \beta(x)$, that is, $y \in \varphi(x)$, thus φ is u.s.c. $\square$.

Berge's Maximum Theorem

Corollary 4.2. If f is a continuous numerical function in $S \times T$ and φ is a continuous mapping of S into T such that, for each x, $\varphi(x) \neq \varnothing$, the numerical function M defined by

$$M(x) = \max\{f(x, y) \mid y \in \varphi(x)\}$$

is continuous in X and the mapping ϕ defined by

$$\phi(x) = \{y \in \varphi(x) \mid f(y) = M(x)\}$$

is u.s.c. of S into T.

Berge's Maximum Theorem

Proof. Define

$$y \prec_x z \qquad \Longleftrightarrow \qquad f(x,y) \prec f(x,z)$$

Since f is continuous, then $\{(x,y,z) \in S \times T \times T \mid f(x,y) \not\prec f(x,z)\}$ is closed, then for any $x \in S$ and $y \in Y$, $\{z \in Y \mid f(x,y) \prec_x f(x,z)\}$ is closed, then f is u.s.c.

We now check that M is continuous. $\{x\}$ is a single-point mapping of correspondence, the product mapping $(x, r(x))$ is u.s.c., f is continuous, then $f(x, r(x))$ is u.s.c., but $f(x, r(x)) = \{M(x)\}$, then $\alpha : S \to \mathbb{R}$ is continuous. $\square$